

20	B	Wire P: $R = \frac{\rho L}{A} = \frac{\rho l}{\pi(d/2)^2} = \frac{4\rho l}{\pi d^2}$ Wire Q: $\frac{1}{4}R = \frac{(\frac{1}{3}\rho)(2l)}{\pi(d_Q/2)^2} = \frac{2}{3} \frac{4\rho l}{\pi d_Q^2}$ $d_Q = 1.6d$ $d^2 = \frac{4\rho l}{\pi R}$ $d_Q^2 = \frac{8}{3} \frac{4\rho l}{\pi R} = \frac{8}{3} d^2$
21	D	$E = 3r + 3(1)$